comprehensive, these serve to illustrate the basic strategies that people have taken.

The virtue of Many-Worlds is in the simplicity of its basic formulation: there is a wave function that evolves according to the Schrödinger equation. All else is commentary. Some of that commentary, such as the split into systems and their environment, decoherence, and branching of the wave function, is extremely useful, and indeed indispensable to matching the crisp elegance of the underlying formalism to our messy experience of the world.

Whatever your feelings might be about Many-Worlds, its simplicity provides a good starting point for considering alternatives. If you remain profoundly skeptical that there are good answers to the problem of probability, or are simply repulsed by the idea of all those worlds out there, the task you face is to modify Many-Worlds in some way. Given that Many-Worlds is just "wave functions and the Schrödinger equation," a few plausible ways forward immediately suggest themselves: altering the Schrödinger equation so that multiple worlds never develop, adding new variables in addition to the wave function, or reinterpreting the wave function as a statement about our knowledge rather than a direct description of reality. All of these roads have been enthusiastically walked down.

We turn first to the possibility of altering the Schrödinger equation. This approach would seem to be squarely in the comfort zone of most physicists; almost before any successful theory has been established, theorists ask how they could play around with the underlying equations to make it even better. Schrödinger himself originally hoped that his equation would describe waves that naturally localized into blobs that behaved like particles when viewed from far away. Perhaps some modification of his equation could achieve that ambition, and even provide a natural resolution to the measurement problem without permitting multiple worlds.

This is harder than it sounds. If we try the most obvious thing, adding new terms like Ψ^2 to the equation, we tend to ruin important features of the theory, such as the total set of probabilities adding up to one. This kind of obstacle rarely deters physicists. Steven Weinberg, who developed the